GGG-C-313C
August 16, 1976
SUPERSEDING
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January 15, 1968

# FEDERAL SPECIFICATION

## CHISELS, HAND: CAPE, DIAMOND POINT, COLD, ROUND NOSE, AND SIDE CUTTING

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

## 1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers hand chisels used for cutting or shaping metals.

1.1.1 Federal specification coverage. Federal specifications do not include all types, classes, and styles of the commodities indicated by the titles of the specifications, but are intended to cover only those most generally used by the Federal Government.

1.2 Classification.

1.2.1 Types, classes, and styles. Hand chisels (metal working) covered by this specification shall be of the following types, classes, and styles, as specified (see 6.2):

Type I. Cape.
  Style A. Normal duty.
  Style B. Heavy duty.

Type II. Diamond point.
  Style A. Normal duty.
  Style B. Heavy duty.

Type IV. Cold
  Class 1. Regular length.
  Class 2. Long.

Type VI. Round nose.
  Class 1. Half.
  Class 2. Full.

Type VII. Side-cutting.

1.2.2 Sizes. Chisels covered by this specification shall be the sizes shown in the respective tables, as specified (see 6.2).

## 2. APPLICABLE DOCUMENTS.

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.



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Federal Specification:

PPP-P-40 - Packaging and Packing of Hand Tools.

Federal Standard:

Fed. Std. No. 358 - Sampling Procedures.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM) Standard:

E 18 - Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials, Tests For.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

3. REQUIREMENTS

3.1 Illustrations. The illustrations shown herein are for the convenience of identification and are not intended to preclude purchase of chisels which are otherwise in accordance with the requirements of this specification.

3.2 Material. Chisels shall be hot-forged of an alloy steel having good cutting, wear, and shock-resisting properties in order to withstand the tests specified in 4.5.

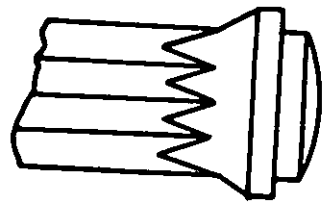
3.3 Hardness.

3.3.1 Work end. Not less than 53 nor more than 59 on a Rockwell C scale when tested as specified in 4.5.1.

3.4 Finish. All surfaces of the chisels shall be free from burrs, roughness, and excessive oxide scale. The ground surfaces forming the cutting edge shall have a bright finish with a maximum of 125 (Arithmetical Average) roughness height rating. The struck end of all chisels shall either be chamfered or provided with an upset head to minimize mushrooming (see optional head design, illustrated below). The top of the struck end shall be finished with either a slight symmetrical crown or at right angles, within 2 degrees, of the longitudinal axis of the body. The chamfer of the struck ends for all chisels, except those having the upset head design, shall be machined or ground within an included angle of 40 to 80 degrees. The length of the chamfer, measured from a flat surface of the body rather than from a corner of the body, shall be a follows:

(a) Chisels 1/4 inch or less across flats; 1/16 inch minimum length.
(b) Chisels 5/16 inch, up to and including 7/16 inch across flats; 3/32 inch minimum length.
(c) Chisels 1/2 inch, up to and including 11/16 inch across flats; 1/8 inch minimum length.
(d) Chisels 3/4 inch and over across flats; 5/32 inch minimum length.

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Optional head design for all chisels striking ends.

3.5 Body shape. The bodies shall be square, hexagonal, or octagonal in cross section. Chisels with a square body shall be either quarter octagon or have the corners suitably rounded.

3.6 Marking. Each chisel shall be plainly and permanently marked with the manufacturer's name or with a trademark of such known character that the source of manufacture may be readily determined.

3.7 Type I, cape. Type I chisels shall have a double-beveled cutting edge ground on both sides to an included angle (see figure 1).

3.7.1 Style A, normal duty. Style A chisels shall conform to table I and be similar to figure 1.

3.7.2 Style B, heavy duty. Style B chisels shall conform to table II and be similar to figure 1.

3.8 Type II, diamond point. Type II chisels shall have the work end tapered and square in cross section. The work end shall be provided with either a single or double bevel having a diamond-shaped surface.

3.8.1 Style A, normal duty. Style A chisels shall conform to table III and be similar to figure 2.

3.8.2 Style B, heavy duty. Style B chisels shall conform to table IV and be similar to figure 2.

3.9 Type IV, cold chisel. Type IV chisels shall have a double-beveled cutting edge ground on both sides to an included angle (see figure 3). The cutting edge shall be straight and shall be 90 degrees to the longitudinal axis of the chisel $\pm$ 2°.

3.9.1 Class 1, regular length. Class 1 chisels shall conform to table V and be similar to figure 3.

3.9.2 Class 2, long. Class 2 chisels shall conform to table VI and be similar to figure 3.

3.10 Type VI, round nose.

3.10.1 Class 1 half. Class 1 chisels shall have a semicircular cutting edge with a single bevel and shall conform to table VII and be similar to figure 4.



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3.10.2 Class 2, full. Class 2 chisels shall have the work end eccentrically tapered with respect to the longitudinal axis of the chisel body. The cutting edge shall be single-bevel ground so that the cutting edge has the shortest radius of the resulting flat, measured from the longitudinal axis of the chisel body. Class 2 chisels shall conform to table VIII and shall be similar to figure 5.

3.12 Type VII, side-cutting. Type VII chisels shall have a single beveled cutting edge. The work end shall be tapered so that the cutting edge is directly in line with the longitudinal axis of the chisel body. Type VII chisels shall conform to table IX and be similar to figure 6.

3.12 Workmanship. The chisels shall be free from rust, fins, burrs, sharp edges, (except cutting edges) or any other defects which may impair their serviceability, durability or appearance.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Inspection of materials and components. In accordance with 4.1 above, the supplier is responsible for insuring that materials and components used were manufactured, tested, and inspected in accordance with the requirements of referenced subsidiary specifications and standards to the extent specified herein, or, if none, in accordance with this specification.

4.2 Sampling procedures. Sampling procedures shall be in accordance with FED-STD-358.

4.3 Testing. Each sample unit shall be tested in accordance with 4.4 thru 4.5.2.2.2.

4.4 Hardness. Hardness tests shall be conducted in accordance with ASTM E-18. When grinding is necessary to prepare the test surface, the amount of material removed must not exceed 0.007 inches on the surface contacted by the indentor.

4.5 Tests.

4.5.1 Hardness test. Chisels shall be subject to a hardness test in accordance with ASTM E-18 to determine compliance with 3.3.1.

4.5.1.1 Work end. Hardness impressions on the work end of the chisels shall be taken within 1/2 inch from the cutting edge and in accordance with ASTM E-18.

4.5.2 Performance test.

4.5.2.1 Specimen for performance test. A square or rectangular carbon-steel bar, in the range of FS 1040 to FS 1060, having a minimum thickness of 3/4 inch and of sufficient length, shall be used in subjecting the chisels to the performance test. The bar shall be secured in a rigid manner on a solid foundation such as an anvil or steel block of not less than 20 pounds. The bar shall show a hardness of not less than 25 nor more than 30 on a Rockwell C scale.



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4.5.2.2 Method of test.

4.5.2.2.1 Types I, IV, & VII chisels. Types I, IV and VII chisels shall be held firmly in as near a vertical position as possible with the work end resting on the flat surface of the test bar. The chisels shall be struck with the applicable machinist's hammer (see table X) as squarely as possible at least 20 blows to produce at least 15 indentations in the test bar.

4.5.2.2.2 Types II, and VI chisels. Types II, and VI chisels shall be held as in 4.5.2.2.1 except at a 45° angle and shall be struck with a machinist's hammer as squarely as possible to produce indentations in the bar. At least 25 indentations shall be made in the test bar by each sample chisel. The size of the hammer used for the various sizes shall be as specified in table XI.

4.5.2.2.3 Any evidence of chipping, spalling, cracking, excessive dulling or turning of the cutting edge; or bending or chipping of the head (struck end) during the test shall be cause for rejection. Note: A minor amount of mushrooming is acceptable.

4.6 Inspection of preparation for delivery requirements. An inspection shall be made to determine that the preservation, packaging, packing, and marking comply with the requirements of PPP-P-40.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing and marking. Preservation, packaging, packing and marking shall be in accordance with PPP-P-40. Preservation, packaging and packing shall be level A, B, or C, as specified (see 6.2).

6. NOTES

6.1 Intended use. Types I, II, IV, VI, and VII chisels are intended for cutting or chipping metals. Caution: Safety goggles should be worn when using these chisels.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type, class, and style of chisel required (see 1.2.1).
(c) Size of chisel (width cutting end) required (see 1.2.2 and applicable tables).
(d) Specify length required for type IV, class 2 chisels (see 3.9.2).
(e) Selection of applicable levels of preservation, packaging, and packing (see 5.1).

6.3 Supersession data. The following chisels which were included in GGG-C-313B have been deleted from GGG-C-313C: Type V, Rivet (rivet buster); Styles A & B under class 1 of Type VI, Round Nose; and Type VIII, Splitting.



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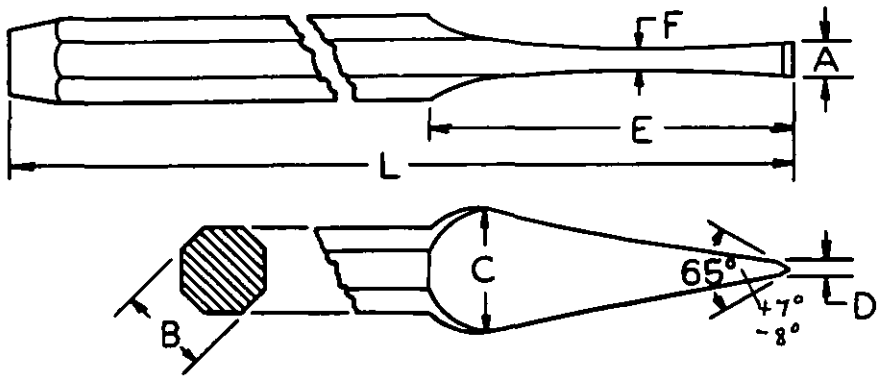



FIGURE 1. Type I, styles A and B, cape, normal duty and heavy duty.

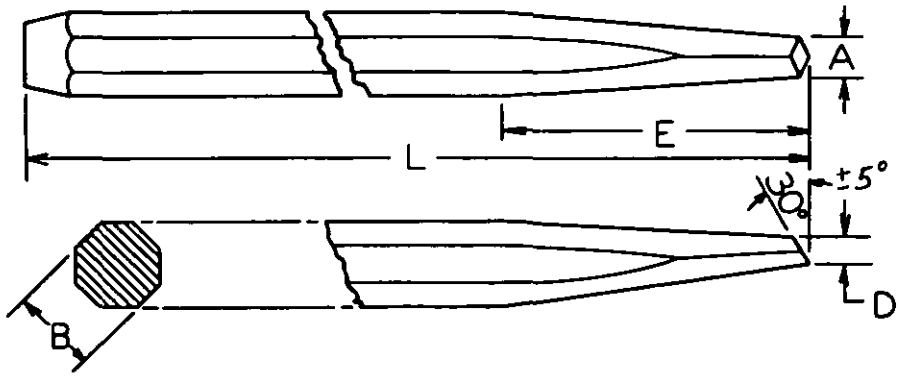



FIGURE 2. Type II, styles A and B, diamond point, normal and heavy duty.

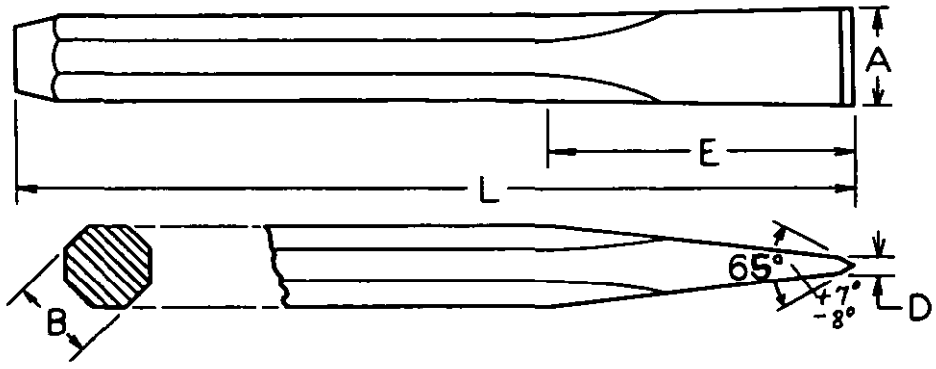



FIGURE 3. Type IV, cold chisel.



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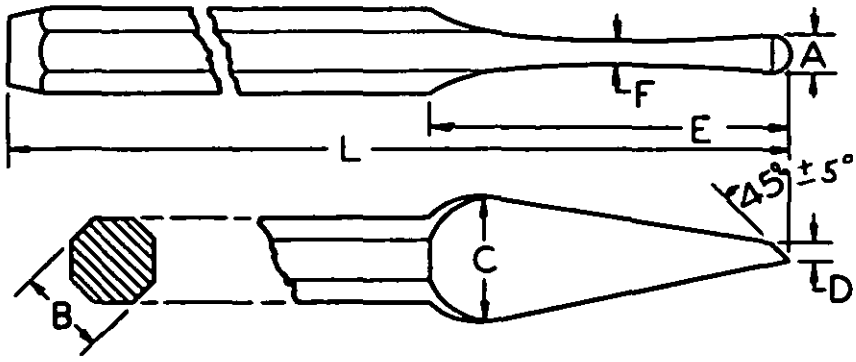



FIGURE 4. Type VI, class 1, half, round nose chisel, normal duty.

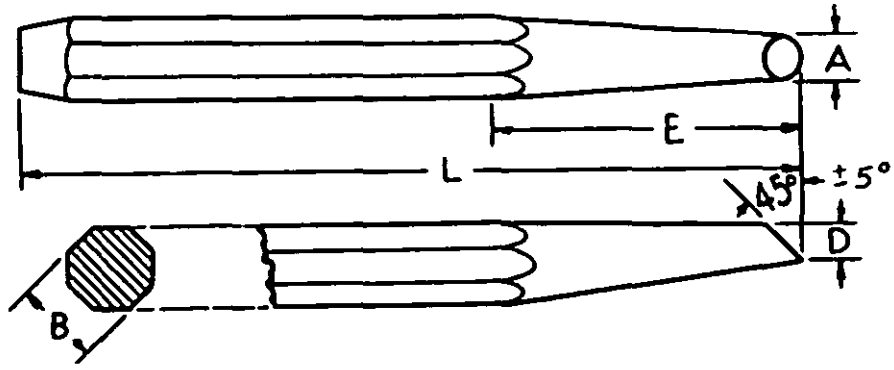



FIGURE 5. Type VI, class 2, full, round nose chisel.

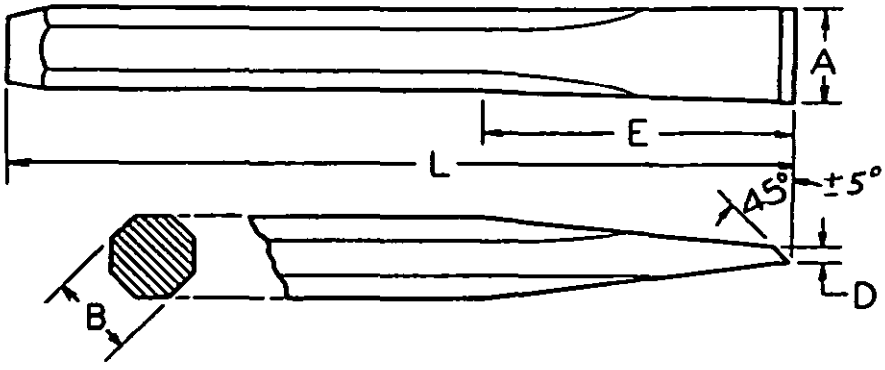



FIGURE 6. Type VII, side-cutting chisel.



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TABLE I. Type I, style A, cape, normal duty

| Size (width of cutting edge) | Distance across flats | Widest section of blade | Thickness at work end | Length | Thinnest section of blade | Overall length |
|---|---|---|---|---|---|---|
| A | B | C | D | E | F | L |
| ± 1/32 | Minimum Stock size | Minimum | ± 1/64 | Minimum | + 1/32 - 1/64 | Minimum |
| Inch | Inch | Inches | Inch | Inches | Inch | Inches |
| 1/8 | 5/16 | 1/2 | 3/32 | 1-1/2 | 3/32 | 4-1/2 |
| 3/16 | 5/16 | 1/2 | 3/32 | 1-5/8 | 1/8 | 5 |
| 1/4 | 3/8 | 9/16 | 3/32 | 1-5/8 | 9/64 | 5 |
| 5/16 | 1/2 | 5/8 | 7/64 | 1-3/4 | 3/16 | 6 |
| 3/8 | 1/2 | 3/4 | 7/64 | 2-1/8 | 1/4 | 6-1/2 |
| 1/2 | 5/8 | 7/8 | 7/64 | 2-3/4 | 11/32 | 7 |
| 5/8 | 3/4 | 15/16 | 1/8 | 3-1/2 | 3/8 | 7 |
| 3/4 | 7/8 | 1-1/6 | 1/8 | 4 | 1/2 | 8 |

TABLE II. Type I, style B, cape, heavy duty

| Size (Width cutting edge) | Distance across flats | Widest section of blade | Thickness at work end | Length | Thinnest section of blade | Overall length |
|---|---|---|---|---|---|---|
| A | B | C | D | E | F | L |
| ± 1/32 | Minimum Stock size | Minimum | ± 1/64 | Minimum | ± 1/64 | Minimum |
| Inch | Inch | Inches | Inch | Inches | Inch | Inches |
| 3/16 | 19/32 | 23/32 | 3/32 | 2 | 1/8 | 6-3/4 |
| 1/4 | 19/32 | 23/32 | 3/32 | 2-5/8 | 5/32 | 6-3/4 |
| 5/16 | 23/32 | 1-3/32 | 1/8 | 2-15/16 | 7/32 | 7-1/4 |

TABLE III. Type II, style A, diamond point, normal duty

| Size (width of cutting edge) | Distance across flats | Thickness at work end | Length of taper | Overall length |
|---|---|---|---|---|
| A | B | D | E | L |
| ± 1/32 | Minimum Stock size | ± 1/32 | Minimum | Minimum |
| Inch | Inch | Inch | Inches | Inches |
| 1/8 | 5/16 | 1/8 | 1-1/2 | 4-1/2 |
| 1/4 | 3/8 | 1/4 | 1-3/4 | 5-1/2 |
| 5/16 | 1/2 | 5/16 | 2 | 5-3/4 |
| 3/8 | 1/2 | 3/8 | 2-1/8 | 6-1/4 |
| 1/2 | 5/8 | 1/2 | 2-3/4 | 7 |
| 5/8 | 3/4 | 5/8 | 3 | 7-1/2 |
| 3/4 | 7/8 | 3/4 | 3 | 8 |



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TABLE IV. Type II, style B, diamond point, heavy duty

| Size (width of cutting edge) A | Distance across flats B | Thickness at work end D | Length of taper E | Overall length L |
|---|---|---|---|---|
| ± 1/32 | Minimum Stock size | ± 1/32 | Minimum | Minimum |
| Inch | Inch | Inch | Inches | Inches |
| 11/32 | 3/4 | 11/32 | 2-1/4 | 7 |

TABLE V. Type IV, class 1, regular length cold chisels

| Size (width of cutting edge) A | Distance across flats B | Thickness of cutting edge D | Nominal length of taper E | Overall length L |
|---|---|---|---|---|
| ± 1/16 - 1/32 | Minimum Stock size | ± 1/64 | Minimum | Minimum |
| Inches | Inch | Inch | Inches | Inch |
| 1/4 | 3/16 | 1/16 | 1-13/32 | 4 |
| 5/16 | 1/4 | 1/16 | 1-1/2 | 4 |
| 3/8 | 5/16 | 1/16 | 1-3/4 | 5 |
| 7/16 | 3/8 | 1/16 | 2 | 5-1/2 |
| 1/2 | 7/16 | 3/32 | 2 | 5-3/4 |
| 5/8 | 1/2 | 3/32 | 2 | 6 |
| 3/4 | 5/8 | 3/32 | 2-1/4 | 6-1/2 |
| 7/8 | 3/4 | 1/8 | 2-1/2 | 7-1/2 |
| 1 | 7/8 | 1/8 | 2-3/4 | 8 |
| 1-1/8 | 1 | 5/32 | 3-1/4 | 8-1/2 |

TABLE VI. Type IV, class 2, long cold chisels

| Size width of cutting edge) A | Distance across flats B | Thickness of cutting edge D | Length of taper E | Overall length L |
|---|---|---|---|---|
| + 1/16 - 1/32 | Minimum Stock size | ± 1/64 | Minimum | ± 1/4 |
| Inches | Inches | Inch | Inches | Inches |
| 5/8 | 1/2 | 3/32 | 2 | 12 |
| 3/4 | 5/8 | 1/8 | 2-1/4 | 8,12,16,18 |
| 7/8 | 3/4 | 1/8 | 2-1/2 | 12,18 |
| 1 | 7/8 | 1/8 | 2-3/4 | 12, 18 |
| 1-1/4 | 1-1/8 | 5/32 | 3-1/4 | 24,36 |
| 1-1/2 | 1-3/8 | 3/16 | 3-1/2 | 24 |



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TABLE VII. Type VI, class 1, normal duty, half round nose chisel

| Size (width of cutting edge) A | Distance across flats B | Widest section of blade C | Thickness at work end D | Length E | Thinnest section of blade F | Overall length L |
|---|---|---|---|---|---|---|
| ± 1/32 | Minimum Stock size | Minimum | ± 1/64 | Minimum | Minimum | Minimum |
| Inch | Inch | Inches | Inch | Inches | Inch | Inches |
| 3/16 | 5/16 | 7/16 | 1/8 | 1-3/4 | 1/8 | 5 |
| 1/4 | 3/8 | 9/16 | 1/8 | 1-3/4 | 3/16 | 5-1/2 |
| 5/16 | 1/2 | 7/8 | 5/32 | 2-1/4 | 3/16 | 6 |
| 3/8 | 1/2 | 7/8 | 3/16 | 2-1/4 | 1/4 | 6 |
| 1/2 | 5/8 | 15/16 | 1/4 | 2-3/4 | 11/32 | 7 |
| 5/8 | 3/4 | 1 | 9/32 | 3-1/4 | 3/8 | 7-1/2 |
| 3/4 | 7/8 | 1-1/6 | 5/16 | 3-1/2 | 15/32 | 8 |

TABLE VIII. Type VI, class 2, full-round nose chisel

| Size (width (dia.) of cutting edge) A | Distance across flats B | Thickness of work end D | Length of taper E | Overall length L |
|---|---|---|---|---|
| ± 1/32 | Minimum Stock size | ± 1/32 | Minimum | Minimum |
| Inch | Inch | Inch | Inches | Inch |
| 1/4 | 3/8 | 9/32 | 2-1/4 | 5-1/2 |
| 5/16 | 1/2 | 11/32 | 2-1/2 | 6 |
| 1/2 | 3/4 | 17/32 | 3 | 7 |

TABLE IX. Type VII, side-cutting chisel.

| Size (width of cutting edge) A | Distance across flats B | Thickness at work end D | Length of taper E | Overall length L |
|---|---|---|---|---|
| ± 1/32 | Minimum Stock size | ± 1/64 | Minimum | Minimum |
| Inch | Inch | Inch | Inches | Inches |
| 7/8 | 23/32 | 1/8 | 2-3/4 | 7 |



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TABLE X, Types I, IV, & VII

| Chisel size - distance across flats<br>Inches | Weight of Hammer<br>oz. |
|---|---|
| 3/16 | 8 |
| 1/4 | 8 |
| 5/16 | 12 |
| 3/8 | 16 |
| 7/16 | 16 |
| 1/2 | 24 |
| 5/8 | 24 |
| 3/4 | 32 |
| 7/8 | 32 |

TABLE XI. Types II and VI

| Width of cutting edge | Hammer | |
|---|---|---|
| | Weight | Handle length |
| Inch | Pounds | Inches |
| up to 7/16 | 1 | 14 |
| 1/2 to 5/8 | 1-1/2 | 15 |
| Over 5/8 | 2 | 16 |

MILITARY INTERESTS:

DOD has waived coordination on revisions and amendments to this Federal Specification until further notice.

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